

1 Temp

There are a homeomorphism $g_q : S^{\lambda_q} \rightarrow D_q/S(q \rightarrow)$ with $g_q(S) = [S(q \rightarrow)]$ and a constant $T_q \geq 0$ such that, with the open neighbourhood

$$W_q := \{x \in D_q \mid \tau(x) > T_q\} \ni q$$

of q (recall $\tau(q) = \infty$), the map g_q restricts to an orientation preserving diffeomorphism from $U_q := g_q^{-1}(W_q)$ onto W_q . In particular, $(D_q, S(q \rightarrow))$, g_q and U_q satisfy the assumptions of Lemma 1.0.1.

Lemma 1.0.1 (Orientation Class of a Disk). *Let $n \geq 1$ and let $(\Sigma, \partial\Sigma)$ be a compact pair such that $\Sigma \setminus \partial\Sigma$ is an oriented C^1 -manifold of dimension n . Let $g : S^n \rightarrow \Sigma/\partial\Sigma$ be a homeomorphism with $g(S) = [\partial\Sigma]$ and let $U \subseteq S^n \setminus \{S\}$ be open such that $g|_U$ is an orientation preserving C^1 -diffeomorphism onto an open subset of $\Sigma \setminus \partial\Sigma$. Define $o_\Sigma := (g^{-1})^*(\beta_n) \in K^{-n}(\Sigma, \partial\Sigma)$. Let V be an oriented n -dimensional vector space and $h : \Sigma/\partial\Sigma \rightarrow \overline{V}$ a pointed map such that for some $z \in V$ we have $h^{-1}(z) = \{y\}$ with $y \in g(U)$, h is C^1 near y and $dh|_y$ is invertible. Then*

$$h^*(\beta_V) = \text{sign det} \left(dh|_y \right) \cdot o_\Sigma.$$

References

- [BH04] A. Banyaga and D. Hurtubise. *Lectures on Morse Homology*. Texts in the Mathematical Sciences. Springer Netherlands, 2004. ISBN: 9781402026959. URL: https://books.google.de/books?id=AX-_sbMjOK4C.
- [Sal90] Dietmar Salamon. “Morse Theory, the Conley Index and Floer Homology”. In: *Bulletin of the London Mathematical Society* 22.2 (Mar. 1990), pp. 113–140. ISSN: 0024-6093. DOI: 10.1112/blms/22.2.113. eprint: <https://academic.oup.com/blms/article-pdf/22/2/113/6692134/22-2-113.pdf>. URL: <https://doi.org/10.1112/blms/22.2.113>.